

Solutions: Calculus Assignment 1

To solve the derivative of the function $f(x) = x^2 + 5x$, we will apply the rules of differentiation provided in the lecture transcript.

Step-by-Step Solution

1. Identify the Function

The given function is:

$$f(x) = x^2 + 5x$$

2. Apply the Sum Rule of Derivatives

The derivative of a sum of functions is the sum of their derivatives. Therefore, we differentiate each term separately:

$$\frac{d}{dx} [f(x)] = \frac{d}{dx} [x^2] + \frac{d}{dx} [5x]$$

3. Differentiate Each Term

According to the lecture transcript:

* For the term x^2 : We use the power rule $\frac{d}{dx} [x^n] = n \cdot x^{n-1}$.

Setting $n=2$:

$$\frac{d}{dx} [x^2] = 2 \cdot x^{2-1} = 2x^1 = 2x$$

* For the term $5x$: We use the rule that the derivative of a constant c multiplied by x (i.e., $c \cdot x$) is c .

Setting $c=5$:

$$\frac{d}{dx} [5x] = 5$$

4. Combine the Results

By adding the derivatives of the individual terms together, we obtain the derivative of the original function:

$$f'(x) = 2x + 5$$

Final Answer

The derivative of the function $f(x) = x^2 + 5x$ is:

$$\mathbf{f'(x) = 2x + 5}$$